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CSE110: Programming Language I (Lab)
Summer 2024
Lab Quiz - 03
Duration: 25 Minutes

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Question 1 [CO2] [10 Points]

Write a Java program that takes two integers **v** and **t** as inputs and assumes **u** as 3.0 and then prints the following outputs based on the inputs provided by the user.

- If **t** is given as less than 1 then print **"Invalid Time"**
- If **v** is divisible by 11 and **t** is an odd number then calculate and print the acceleration **a** in the given format
- Otherwise calculate and print the displacement **s** in the given format

Hints:
v = u + at; where v is final velocity, u is initial velocity, a is acceleration and t is time.
$\mathbf{s = ut + \frac{1}{2}at^2}$; where v is velocity, a is acceleration and s is displacement

Sample Input	Output
Enter the Velocity: 14 Enter the Time: -4	Invalid Time
Enter the Velocity: 22 Enter the Time: 13	The Acceleration is 1.4615384615384615
Enter the Velocity: 20 Enter the Time: 3	The Displacement is 34.5

```
import java.util.Scanner;
public class Exam{
public static void main(String[] args){
 int u = 3.0;
 Scanner sc = new Scanner;
 System.out.println("Enter the velocity:")
 int V = sc.nextInt();
```

```
System.out.println("Enter the time:");
int t =x.nextInt();

if (v%11==0 && t%2!=0){
  double ac =((v-u)/t);
  System.out.println("The Acceleration is"+ac);
}

else if (t<0){
  System.out.println("Invalid time");

}

else {
  double s =(u*t)+(1/2*a*t*t);
  System.out.println("The Displacement is"+s);
  }

}
}
```